

Agile Sprint Planning & User Story Workshop
Smart AI-Based Urban Street Lighting Management System

Course: Software Engineering (CO1 - AT2)

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1. Project Vision & Stakeholders

1.1 Project Vision

To design and deliver an AI-powered, IoT-enabled Urban Street Lighting Management System that continuously monitors streetlight health, automatically detects and reports faults, dynamically optimizes illumination using real-time environmental and traffic data, and predicts maintenance needs - reducing energy costs and manual effort while improving public safety and supporting the city's broader smart-city transformation.

1.2 Project Objectives

- Continuously monitor the operational status of every streetlight in real time.
- Automatically detect faulty or malfunctioning streetlights without manual inspection.
- Use AI to optimize lighting intensity based on traffic, pedestrian, and weather conditions.
- Reduce overall electricity consumption across the city's lighting network.
- Predict maintenance requirements before failures occur.
- Generate energy usage and performance reports for city administrators.
- Improve public safety through reliable, adaptive lighting.
- Support the municipality's smart-city initiatives with centralized, data-driven control.

1.3 Key Stakeholders

Stakeholder	Role / Interest in the Project
Municipal Corporation / City Council	Project sponsor; approves budget, policy, and city-wide rollout.
Urban Planning & Public Works Dept.	Defines infrastructure requirements and integration with road/traffic assets.
Electrical Maintenance Team / Field Technicians	Repairs faulty streetlights based on system alerts; provides field feedback.
IoT Sensor & Hardware Vendors	Supplies light-intensity sensors, controllers, and communication modules.
Product Owner	Owns the product backlog; prioritizes features based on business value.
Scrum Master	Facilitates Agile ceremonies and removes team impediments.
Software Development Team	Designs, builds, and tests the AI/IoT platform and dashboard.

Stakeholder	Role / Interest in the Project
Citizens / General Public	End beneficiaries of safer, better-lit streets; can report faults.
Traffic Police Department	Uses lighting and traffic data to improve road safety enforcement.
Electricity Board / Energy Department	Tracks and bills energy consumption; validates savings claims.
Control Room Operators / System Admins	Monitor the dashboard daily and manage user access and configuration.
Environmental & Smart City Committee	Evaluates sustainability impact and alignment with smart city goals.

2. Epics and User Stories

The product requirements were broken down into 8 Epics, each containing well-structured user stories following the format: “As a <user>, I want <feature>, so that <benefit>.”

EP-1: Real-Time Streetlight Monitoring

Continuous, automated tracking of the operational status of every streetlight across the city.

ID	User Story
US-1	As a control room operator, I want to view the real-time status (ON/OFF/dimmed) of every streetlight on a map, so that I can quickly identify areas experiencing lighting issues.
US-2	As a control room operator, I want the system to continuously collect status data from IoT sensors on each streetlight, so that monitoring is automated and does not depend on manual inspection.
US-3	As a maintenance supervisor, I want to filter streetlights by zone or ward, so that I can monitor and manage specific administrative areas efficiently.

EP-2: Automated Fault Detection & Alerts

Detecting malfunctioning streetlights automatically and notifying the right people instantly.

ID	User Story
US-4	As a control room operator, I want the system to automatically detect a malfunctioning streetlight (no response, flickering, or burnout), so that faults are identified without waiting for citizen complaints.

ID	User Story
US-5	As a maintenance technician, I want to receive instant SMS/app notifications when a fault is detected near me, so that I can respond and repair the issue quickly.
US-6	As a maintenance supervisor, I want faults to be automatically logged with location, timestamp, and fault type, so that repair work can be tracked and audited.

EP-3: AI-Based Adaptive Lighting Control

Using AI to optimize brightness dynamically based on real-world conditions to save energy.

ID	User Story
US-7	As a system administrator, I want the AI engine to automatically adjust brightness based on real-time traffic density and pedestrian movement, so that energy is not wasted lighting empty streets at full brightness.
US-8	As a system administrator, I want lighting intensity to adapt automatically based on weather conditions such as fog or rain, so that visibility and public safety are maintained in poor weather.
US-9	As a energy manager, I want to define minimum and maximum brightness thresholds per zone, so that public safety standards are always met even when the AI dims lights.

EP-4: Predictive Maintenance

Forecasting failures before they happen using historical performance data.

ID	User Story
US-10	As a maintenance supervisor, I want the system to predict which streetlights are likely to fail soon based on historical performance data, so that I can schedule preventive maintenance before an outage occurs.
US-11	As a maintenance planner, I want an automatically generated preventive-maintenance schedule, so that field teams can plan and route their visits efficiently.

EP-5: Energy Usage Analytics & Reporting

Turning consumption data into actionable reports for cost tracking and sustainability goals.

ID	User Story
US-12	As a municipal finance officer, I want to view monthly energy consumption reports by zone, so that I can track cost savings achieved by the smart lighting system.
US-13	As a sustainability officer, I want to compare energy usage before and after AI optimization, so that I can measure and report the environmental impact.

EP-6: Centralized Monitoring Dashboard

A unified control-room view combining status, faults, energy, and maintenance data.

ID	User Story
US-14	As a control room operator, I want a single dashboard showing streetlight status, active faults, energy usage, and maintenance alerts, so that I can manage the entire city's lighting from one screen.
US-15	As a city administrator, I want role-based access to the dashboard, so that different stakeholders see only the information relevant to their role.

EP-7: Citizen Engagement & Complaint Management

Giving citizens a channel to report issues the sensors may have missed.

ID	User Story
US-16	As a citizen, I want to report a faulty streetlight through a mobile app with photo and location, so that the issue gets fixed quickly even if a sensor missed it.
US-17	As a citizen, I want to track the status of my streetlight complaint, so that I know when the repair is expected to be completed.

EP-8: System Administration & Security

Managing users, roles, access, and system configuration safely across the platform.

ID	User Story
US-18	As a system administrator, I want to manage user accounts and access permissions, so that only authorized personnel can view or control lighting parameters.

ID	User Story
US-19	As a system administrator, I want to remotely configure and update AI/lighting control parameters, so that adjustments can be made without physically visiting each streetlight controller.
US-20	As a security officer, I want all critical actions (user changes, manual overrides) to be recorded in an audit trail, so that the system remains secure, traceable, and accountable.

3. Acceptance Criteria

Acceptance criteria for the highest-priority user stories are defined below using the Given-When-Then format. Each row represents one testable scenario.

US-1: View real-time streetlight status on a map

Given	When	Then
Given The operator opens the monitoring dashboard	When the system loads status data	Then every streetlight appears as a marker colored by state (green=ON, red=OFF, yellow=dimmed) within 5 seconds
Given A streetlight's status changes	When the sensor reports the new state	Then the map marker updates within 30 seconds without a manual page refresh

US-2: Continuous IoT status data collection

Given	When	Then
Given A streetlight controller is powered on	When it connects to the network	Then it transmits status data to the cloud platform at a fixed interval (e.g., every 60 seconds)
Given A sensor fails to report for 3 consecutive intervals	When the system checks connectivity	Then the streetlight is flagged as 'communication lost' rather than silently dropped

US-3: Filter streetlights by zone/ward

Given	When	Then
Given The operator selects a ward from the filter panel	When the filter is applied	Then only streetlights in that ward are shown on the map and in the list view

US-4: Automatic fault detection

Given	When	Then
Given A streetlight reports zero light output while scheduled ON	When the fault-detection engine evaluates the reading	Then the light is marked 'Faulty - No Output' within 2 minutes
Given A streetlight's brightness fluctuates beyond a defined threshold repeatedly	When the engine detects the pattern	Then it is flagged as 'Flickering' and added to the fault list

US-5: Instant fault notifications to technicians

Given	When	Then
Given A new fault is confirmed by the system	When a technician is assigned to that zone	Then an SMS and app push notification are sent within 1 minute containing location and fault type

US-7: AI-based adaptive brightness using traffic/pedestrian data

Given	When	Then
Given Traffic and pedestrian sensors report low activity on a street segment	When the AI engine evaluates current conditions	Then brightness is reduced to the pre-configured energy-saving level within safety limits
Given Traffic or pedestrian activity increases suddenly	When the AI engine detects the change	Then brightness is restored to full/safe level within 10 seconds

US-14: Centralized monitoring dashboard

Given	When	Then
Given An authorized user logs into the dashboard	When the home screen loads	Then status summary, active fault count, energy usage, and pending maintenance alerts are all visible on one screen

US-16: Citizen fault reporting via mobile app

Given	When	Then
Given A citizen finds a faulty streetlight	When they submit a report with a photo and GPS location through the app	Then a new complaint ticket is created and routed to the maintenance queue within 1 minute

4. Product Backlog Prioritization

The full product backlog below is prioritized by business value, dependency order, and customer impact, using a High / Medium / Low priority scheme.

Story ID	Epic	Priority	Points	Prioritization Rationale
US-1	EP-1	High	5	Foundational - all other features depend on live status data
US-2	EP-1	High	8	Core IoT data pipeline; blocks monitoring, faults, and AI control
US-4	EP-2	High	8	Directly solves the core problem: delayed fault identification
US-5	EP-2	High	3	High business value; fast technician response reduces downtime
US-7	EP-3	High	13	Primary energy-saving objective and AI differentiator
US-14	EP-6	High	8	Key deliverable for centralized city-wide operations
US-3	EP-1	Medium	3	Improves usability but not blocking for MVP
US-6	EP-2	Medium	3	Needed for audit trail; depends on US-4
US-8	EP-3	Medium	8	Enhances safety; depends on AI engine from US-7

Story ID	Epic	Priority	Points	Prioritization Rationale
US-9	EP-3	Medium	3	Safety guardrail; depends on US-7
US-10	EP-4	Medium	13	High long-term value but needs historical data first
US-12	EP-5	Medium	5	Important for stakeholder reporting; depends on US-2 data
US-16	EP-7	Medium	5	Improves coverage/public trust; independent feature
US-11	EP-4	Low	5	Depends on US-10 predictions being available first
US-13	EP-5	Low	3	Nice-to-have analytical comparison; depends on US-12
US-15	EP-6	Low	3	Security enhancement; can follow after core dashboard (US-14)
US-17	EP-7	Low	2	Minor UX addition on top of US-16
US-18	EP-8	Low	3	Administrative; needed before wide rollout, not for MVP demo
US-19	EP-8	Low	5	Useful operational convenience; depends on AI engine (US-7) existing first
US-20	EP-8	Low	3	Security/compliance requirement; needed before wide rollout, not for MVP demo

5. Sprint Backlog (Sprint 1)

Sprint 1 focuses on the foundational monitoring and fault-detection capabilities, since every later feature (AI optimization, predictive maintenance, dashboards) depends on reliable real-time status data.

5.1 Selected User Stories

ID	User Story	Points
US-1	As a control room operator, I want to view the real-time status (ON/OFF/dimmed) of every streetlight on a map, so that I can quickly identify areas experiencing lighting issues.	5

ID	User Story	Points
US-2	As a control room operator, I want the system to continuously collect status data from IoT sensors on each streetlight, so that monitoring is automated and does not depend on manual inspection.	8
US-3	As a maintenance supervisor, I want to filter streetlights by zone or ward, so that I can monitor and manage specific administrative areas efficiently.	3
US-4	As a control room operator, I want the system to automatically detect a malfunctioning streetlight (no response, flickering, or burnout), so that faults are identified without waiting for citizen complaints.	8
US-5	As a maintenance technician, I want to receive instant SMS/app notifications when a fault is detected near me, so that I can respond and repair the issue quickly.	3

Total Sprint 1 commitment: 27 story points (assuming a team velocity of ~25-30 points per 2-week sprint).

5.2 Task Breakdown

Story ID	Development Task
US-1	Design map-based UI wireframe for streetlight status view
US-1	Integrate mapping library (e.g., GIS/maps API) with backend status feed
US-1	Implement color-coded marker logic (ON/OFF/dimmed) and auto-refresh
US-2	Define IoT communication protocol (MQTT/HTTP) between controllers and cloud
US-2	Build ingestion service to receive and store sensor status data
US-2	Implement 'communication lost' detection for non-reporting devices
US-3	Add ward/zone metadata to streetlight records
US-3	Build filter UI component and backend query support
US-4	Design rule-based/AI logic to classify 'no output' and 'flickering' faults
US-4	Implement fault classification service and store fault records
US-4	Unit test fault detection against simulated sensor data
US-5	Integrate SMS/push notification gateway
US-5	Build technician-zone assignment mapping
US-5	Trigger notification workflow on new confirmed fault

6. Story Point Estimation

Story points were estimated using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13), based on complexity, effort, and uncertainty relative to a baseline story (US-3, filter by zone = 3 points, treated as a simple, well-understood feature).

Story ID	Points	Justification
US-1	5	High priority; Foundational - all other features depend on live status data
US-2	8	High priority; Core IoT data pipeline; blocks monitoring, faults, and AI control
US-4	8	High priority; Directly solves the core problem: delayed fault identification
US-5	3	High priority; High business value; fast technician response reduces downtime
US-7	13	High priority; Primary energy-saving objective and AI differentiator
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US-8	8	Medium priority; Enhances safety; depends on AI engine from US-7
US-9	3	Medium priority; Safety guardrail; depends on US-7
US-10	13	Medium priority; High long-term value but needs historical data first
US-12	5	Medium priority; Important for stakeholder reporting; depends on US-2 data
US-16	5	Medium priority; Improves coverage/public trust; independent feature
US-11	5	Low priority; Depends on US-10 predictions being available first
US-13	3	Low priority; Nice-to-have analytical comparison; depends on US-12
US-15	3	Low priority; Security enhancement; can follow after core dashboard (US-14)
US-17	2	Low priority; Minor UX addition on top of US-16
US-18	3	Low priority; Administrative; needed before wide rollout, not for MVP demo

Story ID	Points	Justification
US-19	5	Low priority; Useful operational convenience; depends on AI engine (US-7) existing first
US-20	3	Low priority; Security/compliance requirement; needed before wide rollout, not for MVP demo

6.1 Estimation Notes

- 1-3 points: Simple UI or CRUD-style features with little integration risk (e.g., filters, complaint tracking).
- 5 points: Moderate complexity involving a single external integration (e.g., mapping, notifications, reporting).
- 8 points: Higher complexity involving multiple integrations or aggregation logic (e.g., IoT ingestion, dashboards, weather-based control).
- 13 points: High complexity/uncertainty involving AI or ML models and multiple data sources (e.g., adaptive traffic-based lighting, predictive maintenance).

7. Sprint Goal & Expected Deliverables

7.1 Sprint Goal

“Deliver a working real-time streetlight monitoring and automated fault-detection system, so that control room operators can view live status on a map and maintenance technicians are alerted instantly to faults, without depending on manual inspection.”

7.2 Expected Sprint Deliverables

- A functional map-based dashboard showing real-time ON/OFF/dimmed status for all streetlights, filterable by zone/ward.
- A working IoT data-ingestion pipeline collecting streetlight status at regular intervals, with 'communication lost' detection.
- An automated fault-detection engine that identifies 'no output' and 'flickering' faults and logs them with location and timestamp.
- An SMS/app notification workflow that alerts the assigned technician within 1 minute of a confirmed fault.
- A demo-ready increment showing end-to-end flow: sensor data → status update → fault detection → technician alert.

7.3 Definition of Done

- All Sprint 1 user stories meet their documented acceptance criteria.
- Code is reviewed, unit tested, and merged to the main branch.

- The dashboard and notification flow are demonstrated in the Sprint Review.
- No critical or high-severity defects remain open.